

Home Services, Scheduled Calmly. Dispatched Instantly.

Cross-platform mobile application built with React Native, Expo SDK 57, and TypeScript.
Connecting homeowners with verified local home-service professionals via intelligent cloud scheduling and real-time messaging.

AUTHOR / DEVELOPER

Maksym Shepeta

CLOUD BACKEND

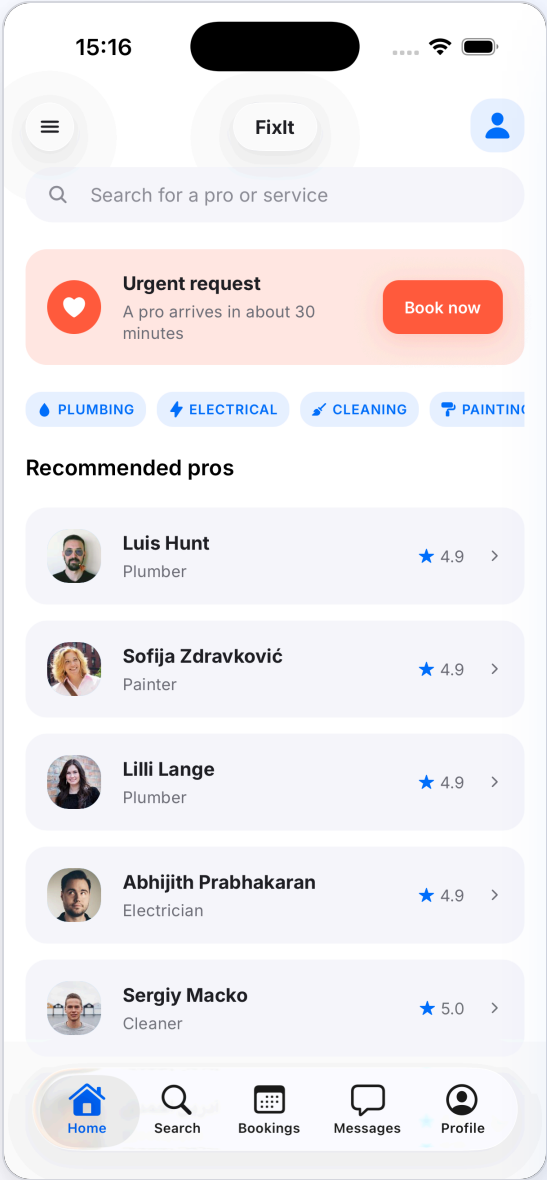
Firebase Auth & Firestore

TECHNOLOGY FOUNDATION

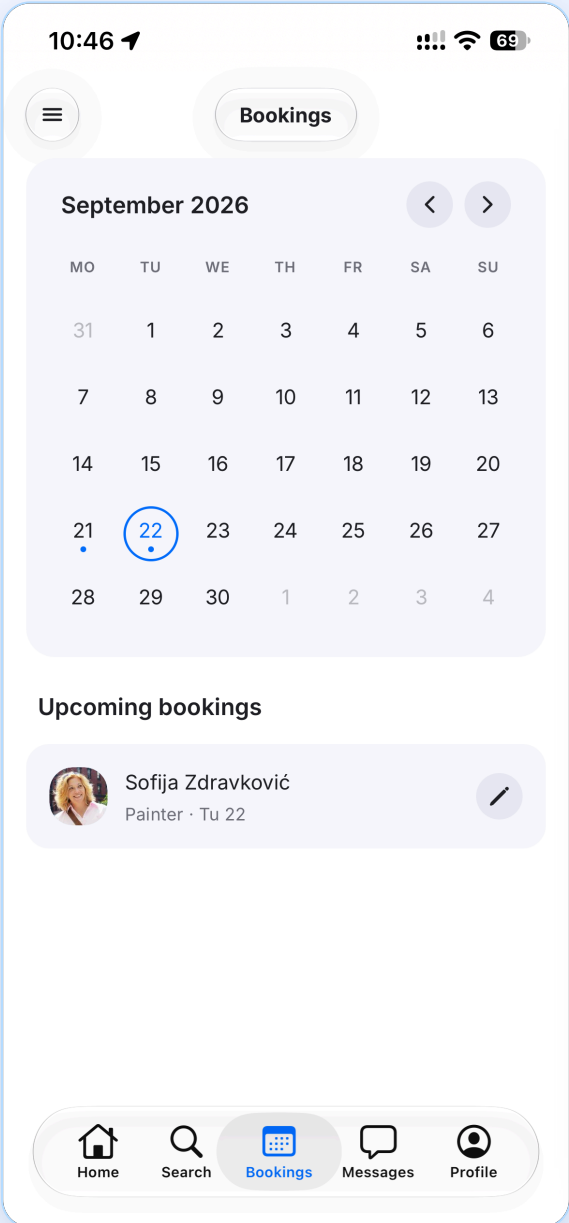
React Native 0.86 · Expo 57

STATE MANAGEMENT

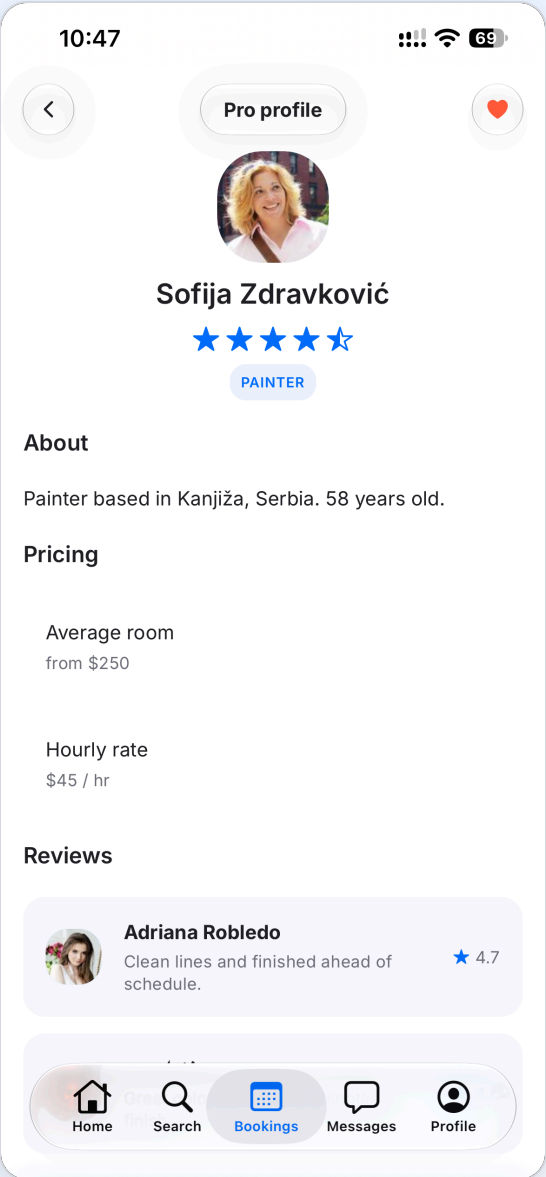
Redux Toolkit + Context API



Home Marketplace



Month Calendar



Verified Pro Profile

The Problem & The FixIt Value Proposition

Bridging homeowners and verified contractors through transparent pricing and instant dispatch

The Homeowner Frustration

Home repairs are historically unpredictable: finding certified tradespeople takes hours of calling, quotes are opaque, arrival times are unreliable, and emergencies require immediate 30-minute intervention rather than week-long waiting lists.

The FixIt Solution: Dual Service Modes

- Scheduled Appointments:** Transparent flat rates and hourly quotes across 6 core trades (Plumber, Electrician, Cleaner, Painter, Carpenter, Gardener).
- Urgent 30-Minute Dispatch:** One-tap emergency callout engine instantly finding active, available specialists today.
- Curated Quality & Ratings:** Real customer reviews, star ratings, and vetted professional credentials.
- Direct 1:1 In-App Messaging:** Real-time Firestore chat threads to coordinate details, clarify tasks, and confirm arrival.

Plumbing

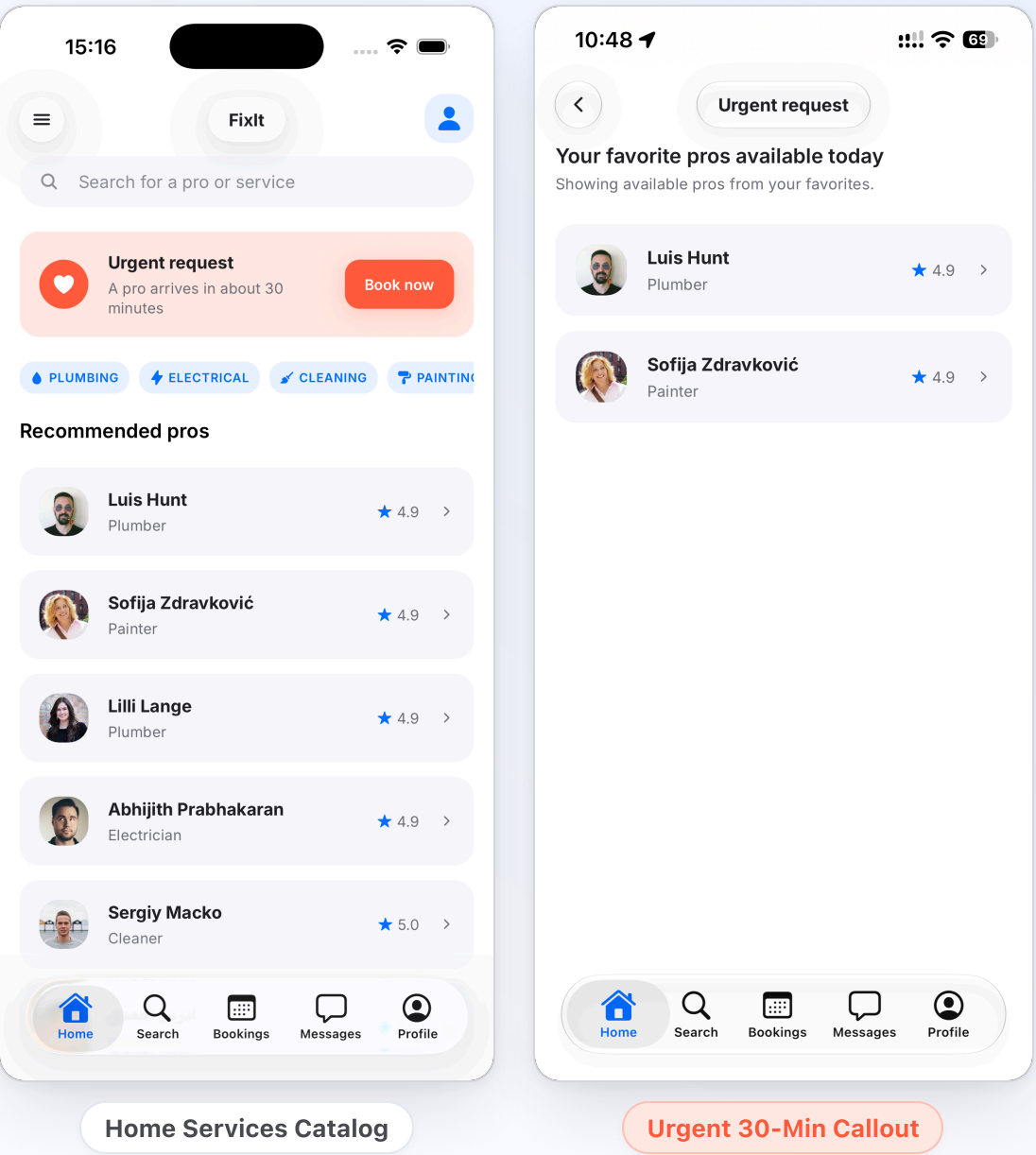
Electrical

Cleaning

Painting

Carpentry

Gardening



From Baseline Prototype to Production MVP

Rigorous assessment of initial homework shortcomings and strategic upgrades implemented

BASELINE VERSION

Initial Prototype Capabilities & Deficits

What worked initially:

- Basic UI screens for Home, Categories, and Profile.
- Demo provider catalog fetched from seeded `randomuser.me`.
- Preliminary tab bar and drawer navigation.

Identified Critical Gaps:

- **No Cloud Persistence:** All bookings and preferences lived in ephemeral memory and vanished on reload.
- **No User Identity:** No registration, login, or user data security.
- **Static Booking:** No interactive calendar; no busy/free day indicators; no reschedule flow.
- **Missing Messaging:** No communication channel between customer and contractor.
- **UI Jerkiness:** Tab bar chrome flickered; edge gestures collided with Drawer.

PRODUCTION MVP

Strategic Enhancements Executed

3 Key Upgrade Vectors:

- **Vector 1: Cloud Backend & Auth (Firebase):** Complete Firebase Auth, hardware Keychain session storage via `expo-secure-store`, and real-time multi-user Firestore persistence.
- **Vector 2: Full-Month Calendar Engine:** Custom interactive `MonthCalendar` with booking dot indicators, date filtering, and reschedule availability modal.
- **Vector 3: Realtime Chat & Favorites:** 1:1 Firestore messaging with auto-replies, keyboard handling, and pro favoriting filtering urgent requests.
- **Native iOS Polish:** 60/120fps native transitions, interactive swipe-back gesture harmony, and Reanimated cascading reveals.

Outcome: Transitioned from an in-memory test script into a complete, usable, commercially-viable mobile product.

State Architecture: Context API vs. Redux Toolkit

Deliberate separation of presentation preferences from asynchronous business domain data

UI PRESENTATION LAYER

React Context API (ThemeContext)

Dedicated to global visual styling parameters and appearance synchronization:

- **Low Mutation Frequency:** Theme mode (Light / Dark / System) changes infrequently — only upon deliberate user toggle or device OS mode shift.
- **Direct Native Bridge:** Injects resolved palette directly into React Navigation's root theme prop and invokes `Appearance.setColorScheme()` to sync Liquid Glass materials natively.
- **Zero Reducer Boilerplate:** Eliminates action creators, dispatchers, and serialization overhead for a simple global setting.
- **Clean Decoupling:** Theme switches never trigger database reads or business slice recalculations.

ThemeContext.tsx

AppearanceScreen.tsx

GlassSurface & Palettes

BUSINESS DOMAIN LAYER

Redux Toolkit (Central Store)

Engineered for high-frequency, relational, asynchronous cloud business logic:

- **Asynchronous Thunk Lifecycles:** `createAsyncThunk` standardizes pending, fulfilled, and rejected handling for Firebase Auth, Firestore queries, and retries.
- **Relational Domain State:** Manages 4 interdependent slices: `authSlice` (Keychain tokens), `bookingsSlice` (active appointments), `chatsSlice` (1:1 messages), and `favoritesSlice`.
- **Selective Subscriptions (`useAppSelector`):** Prevents global re-renders. Updating a single chat message or toggling a favorite heart only re-renders that component, maintaining 60/120fps.
- **Single Source of Truth:** Serializable state tree enables predictable debugging and zero data drift.

authSlice

bookingsSlice

chatsSlice

favoritesSlice

Cloud Authentication & Encrypted Profile

Hardware-encrypted Keychain session storage, Firebase Auth, and native Liquid Glass inputs

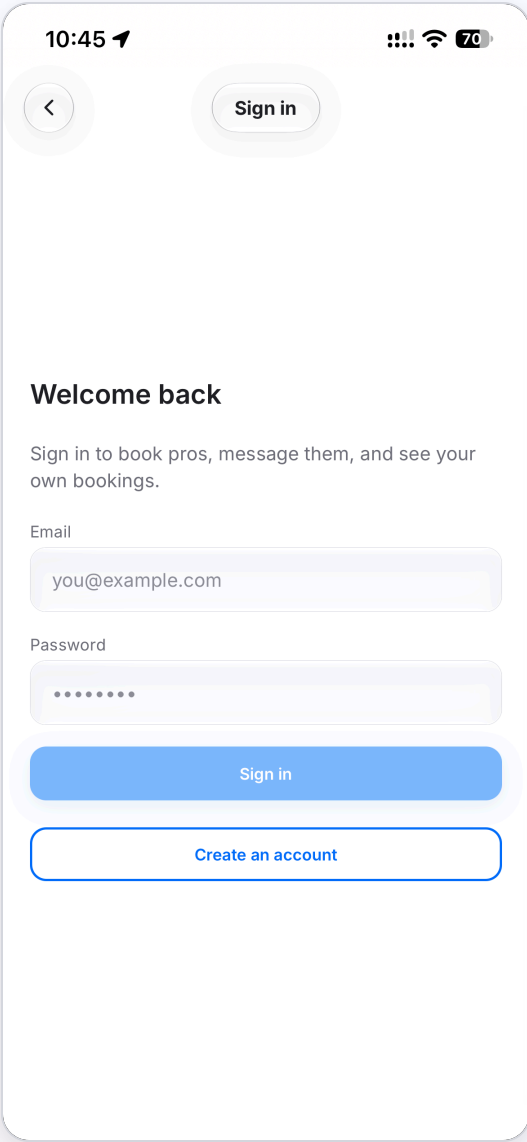
Identity & Session Security

- Complete user onboarding flow connecting customers to cloud accounts with full data privacy.
- Firebase Authentication:** Email/password registration, secure credential hashing, session token refresh, and sign-out.
 - Hardware Keychain Security:** Implements expo-secure-store with hex-sanitized key encoding to store active auth tokens securely on iOS Keychain / Android Keystore.
 - Liquid Glass Input Ergonomics:** Unified TextField and gender selector sharing 48px height, rounded corners, subtle glass border, and vertical text centering.
 - Native iOS Modal Selector:** Custom gender selector presenting clean checkmark selection and instant dismiss.

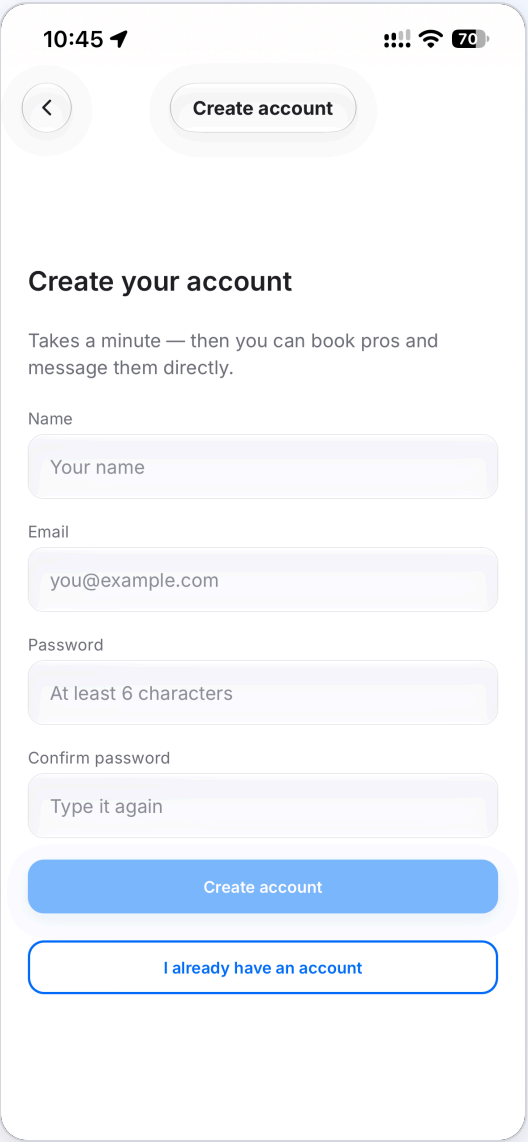
src/api/auth.ts

src/store/authSlice.ts

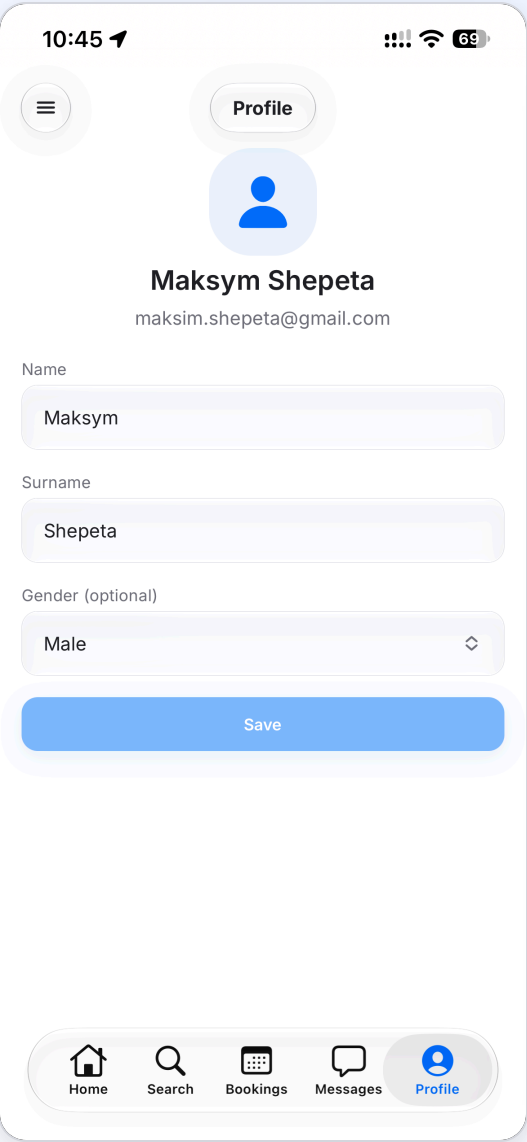
src/screens/ProfileScreen.tsx



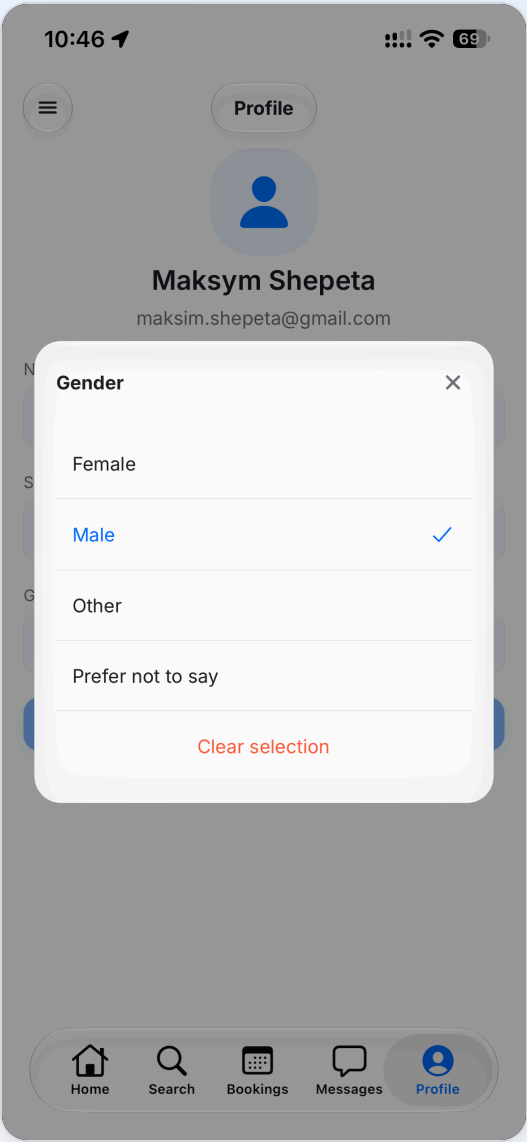
Sign In



Sign Up



Profile Details



Gender Picker

Interactive Month Calendar & Booking Engine

Full-month calendar with visual occupancy dots, dual schedule filtering, and rescheduling modal

Smart Calendar Architecture

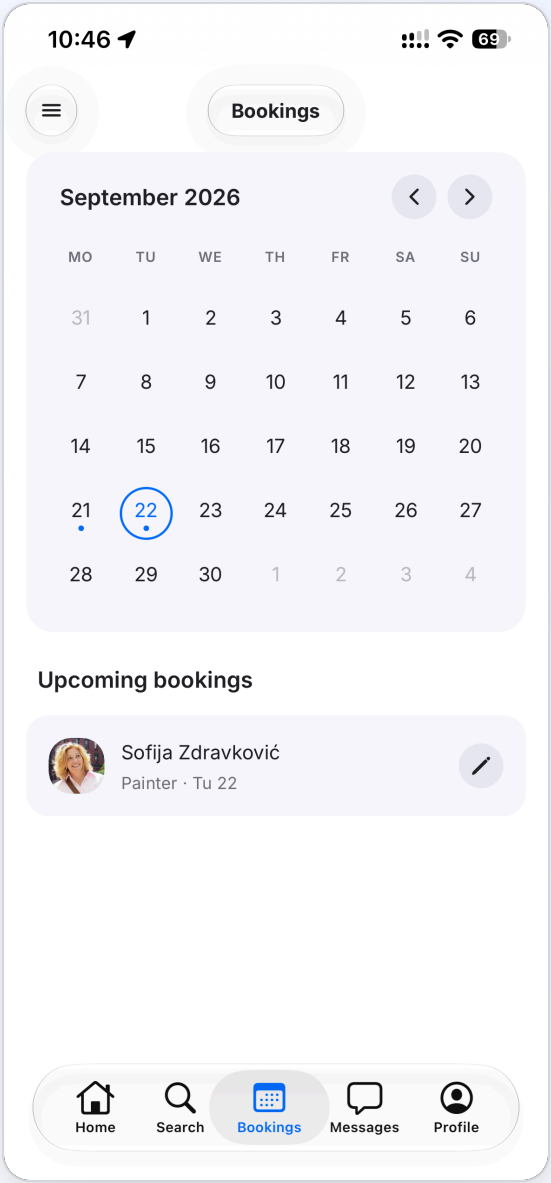
A comprehensive scheduling dashboard giving homeowners instant visibility into active home services.

- **Interactive Full-Month Grid:** Custom-engineered MonthCalendar component supporting smooth month-to-month navigation with adjacent month days seamlessly filling the grid.
- **Visual Occupancy Dots:** Blue point indicators highlight days containing active appointments for instant visual recognition.
- **Dual-View Schedule List:** Displays the top 3 upcoming bookings by default; tapping any specific calendar date immediately filters to appointments for that day.
- **Instant Rescheduling:** Integrated AvailabilityCalendar modal allowing users to pick new available dates with automated capacity checks (max 3 bookings per pro/day).
- **Firestore Cloud Sync:** Real-time persistence under the bookings collection with cancellation support.

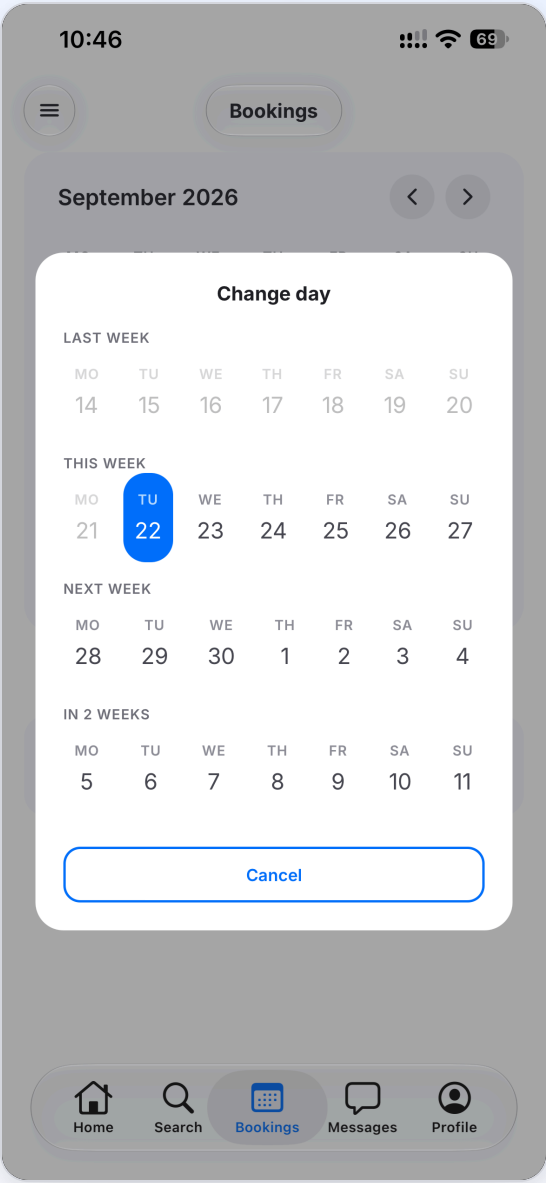
src/components/MonthCalendar/

src/screens/BookingsScreen.tsx

src/api/bookings.ts



Month Calendar & Upcoming List



Reschedule Availability Modal

Direct Messaging & Curated Favorites

1:1 real-time Firestore chat threads and fast-track urgent emergency dispatch

Customer-Pro Connection

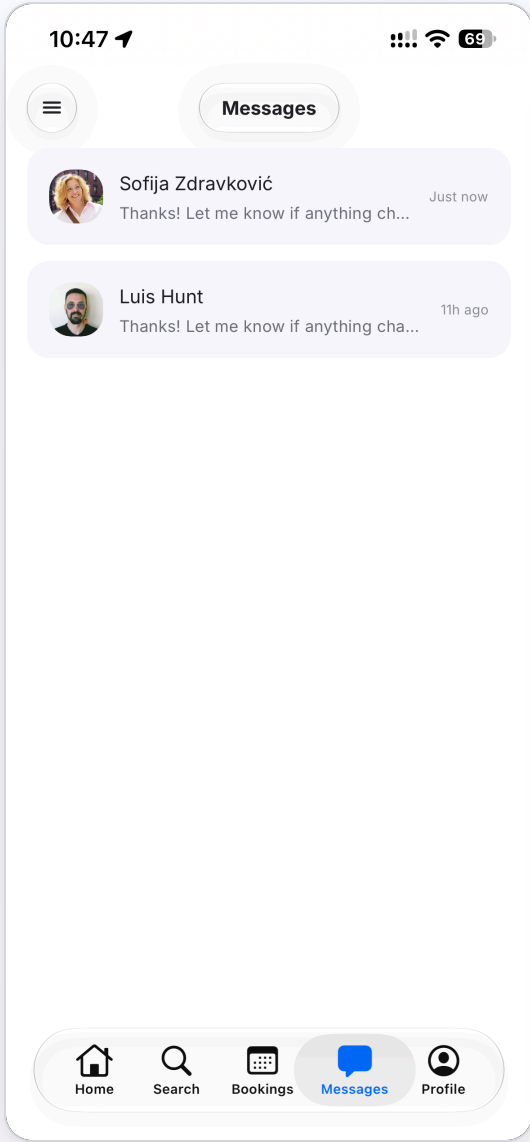
Connecting customers directly to specialists for quote clarification and rapid emergency assistance.

- **1:1 Real-Time Chat:** Threaded conversation channels stored under conversations and messages in Firestore, tied to unique (userId, proId) pairs.
- **Automated Contractor Replies:** Simulated technician responses confirming availability and appointment details.
- **Dynamic Keyboard Handling:** Seamless iOS keyboard avoidance eliminating unsightly white space or buried input bars.
- **Curated Favorites System:** Tap the Liquid Glass heart in the pro header to bookmark contractors. Favorites dynamically filter the Urgent 30-min Callout screen.

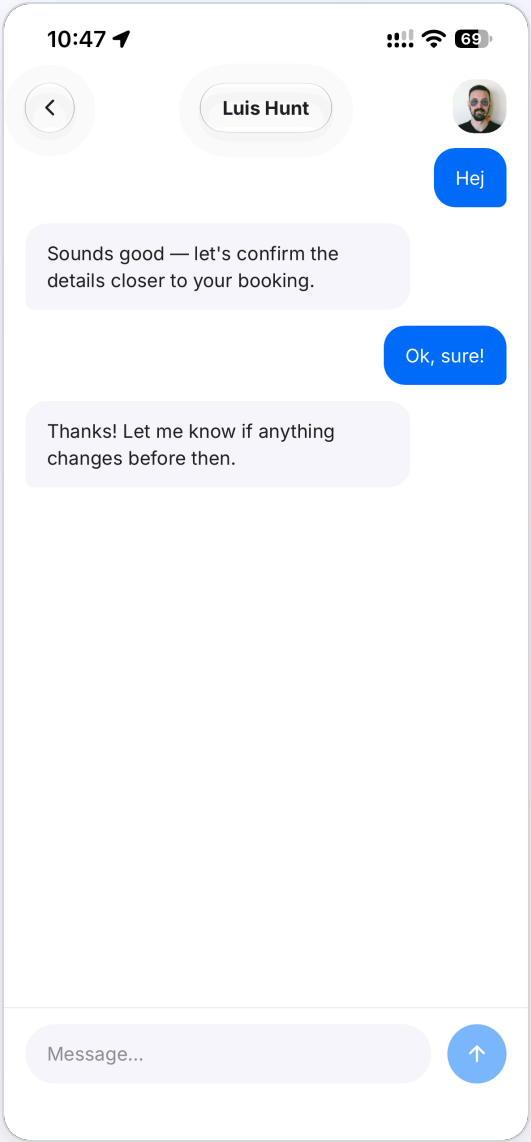
src/screens/ChatScreen.tsx

src/api/chats.ts

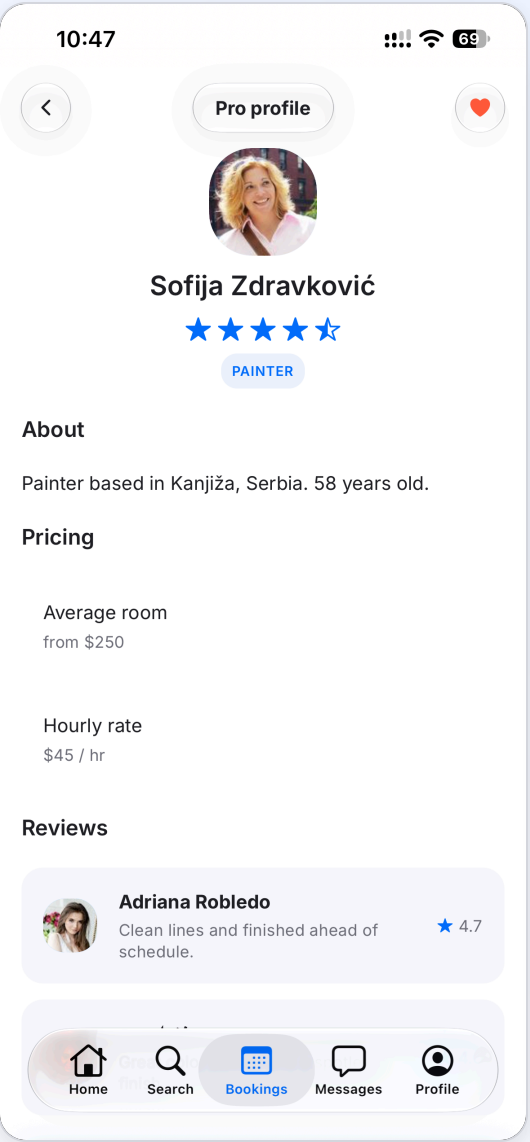
src/store/favoritesSlice.ts



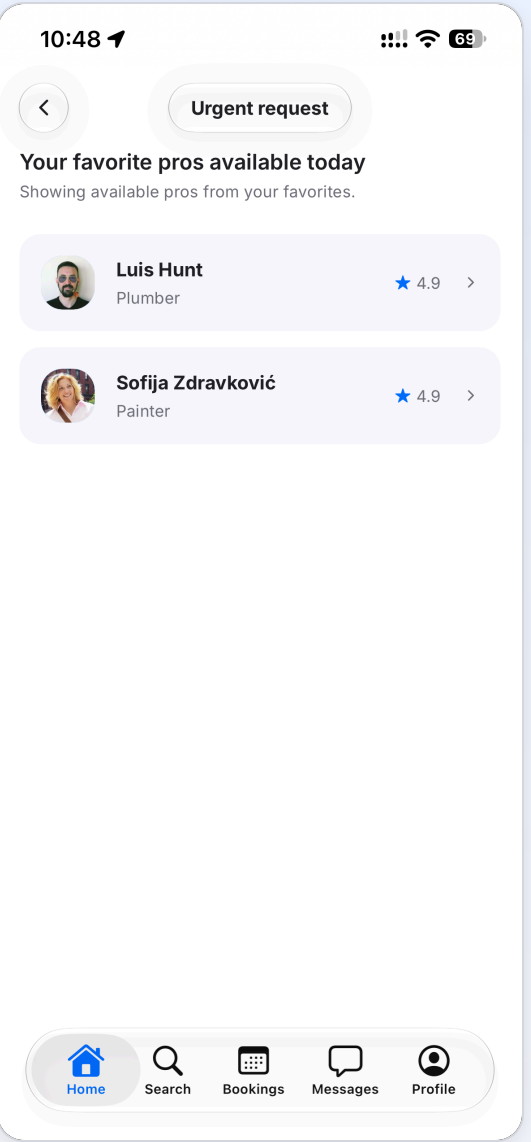
Messages Inbox



Active 1:1 Chat



Header Heart Action



Urgent Favorites Filter

Native iOS 27 Liquid Glass & Fluid Navigation

High-fidelity 60/120fps transitions, Reanimated cascading motion, and adaptive responsive grid

Design System & Platform Polish

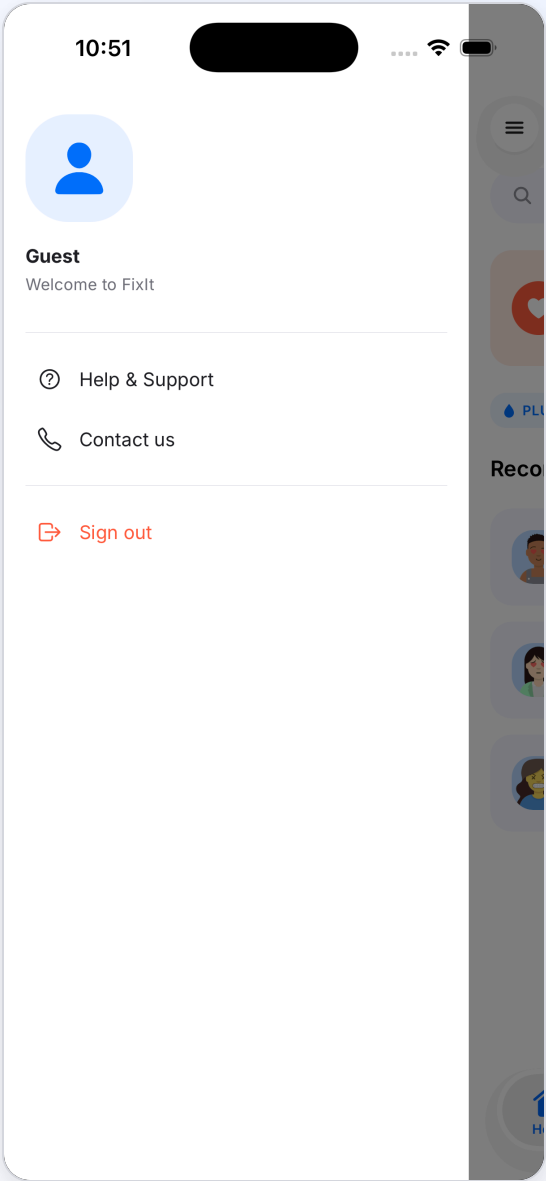
Engineered specifically to feel like a first-party iOS application with silky smooth hardware-accelerated animations:

- **Real Liquid Glass UI (GlassSurface):** Translucent glass materials on iOS 26+ devices via expo-glass-effect with graceful blur fallback.
- **Native Stack 60/120fps Transitions:** Modal and detail screens push with native slide curves; edge swipe gestures operate in harmony with Drawer navigation without conflict.
- **Reanimated Cascading Animations:** Staggered `FadeInDown.delay()` reveals on Home, Profile, Pro details, and Help FAQ accordions.
- **Automatic Tab Bar Hiding:** Bottom navigation bar automatically retreats when entering chat threads or authentication screens.
- **Adaptive Responsive Grid:** Single-column phone portrait reflows dynamically to a 2-column grid on tablets or landscape orientation.

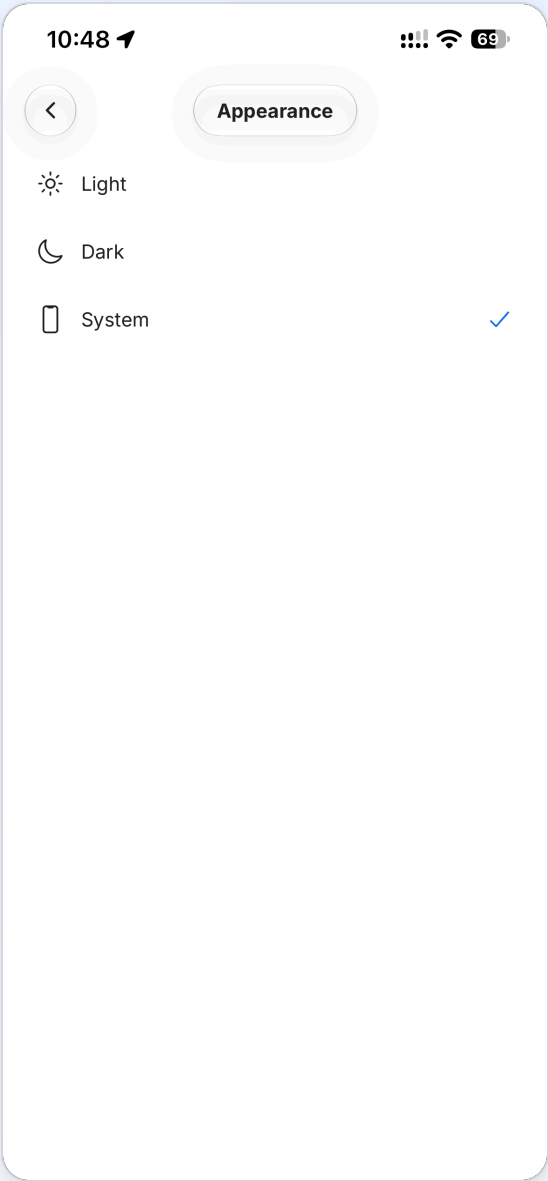
src/components/GlassSurface/

src/navigation/RootNavigator.tsx

src/hooks/useResponsiveLayout.ts



Native Drawer Menu



Appearance (Theme Picker)

Engineering Quality & Submission Verification

Strict compliance with final project rubric and high software engineering standards

0 Errors

TypeScript Strict Mode

Clean compilation via `tsc --noEmit` across entire codebase.

0 Warnings

ESLint Standards

Conforms to Expo 57 and React Native rules with 0 warnings.

100% Modularity

Component Architecture

Single-responsibility component files, separate API, slices, and theme tokens.

Project Ready for Final Evaluation

FixIt mobile application has been thoroughly developed, tested on a physical iPhone 13 Pro (iOS 27), and fully documented.

GIT REPOSITORY

github.com/mikk1max/goit-react-native-hw

Developer: Maksym Shepeta